Debt Capacity and the Anatomy of Sovereign Crises

Sangdong Kim
Korea Insurance Research Institute

Gabriel Mihalache
The Ohio State University



November 2026

Sovereign defaults are rarely complete

- Governments in distress keep paying some obligations and suspend others.
- Missed payments accumulate as arrears that add to the debt.
- Countries often leave a default episode owing more than when it began.

Arellano, Mateos-Planas, and Ríos-Rull (2023): a theory of partial default.

- The sovereign chooses what share of its debt service to withhold.
- It never loses market access. The price reflects the expected default intensity.
- Data: in partial default a third of the time, withholding $\sim$40% of debt service.

Partial default is the right description of sovereign distress, and the natural baseline.

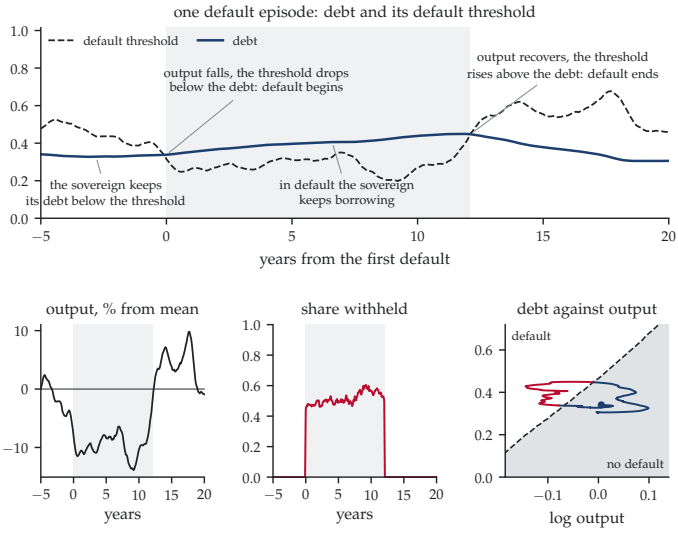
What does partial default teach us about sovereign crises?

Four lessons.

1. Debt capacity. Set by the cost of defaulting, not the value of market access.
2. Entry. Defaults begin with a jump, when shocks overwhelm a precautionary buffer.
3. The episode. A financing regime: borrowing continues until output recovers.
4. Pessimism. Lenders' expectations alone can produce a slow crisis, without a run.

Three versions of the model: deterministic, a Markov chain, and a calibrated diffusion.
Tractable, not stylized: the calibrated model fits the data about as well as theirs.

One default episode in the calibrated model



Why continuous time

- The default intensity is a flow decision. Arrears accrue continuously.
- The sovereign never loses access: no stopping problem, no variational inequality.
- One HJB equation and one linear pricing equation, at every state.
- Given the price, the default decision is static: closed-form thresholds.
- Deterministic model: a pair of ODEs, with closed forms at the steady state.
- Computation: no smoothing shocks, exact moments and impulse responses.

Related literature

- **Partial default and arrears:** Arellano, Mateos-Planas, and Ríos-Rull (2023).
- **Binary default:** Eaton–Gersovitz (1981), Aguiar–Gopinath (2006), Arellano (2008).
- **Long-term debt:** Hatchondo–Martinez (2009), Chatterjee–Eyigungor (2012).
- **Dilution:** Hatchondo–Martinez–Sosa-Padilla (2016), Aguiar–Amador (2020).
- **Continuous time:** Bornstein (2020), Tourre (2017), Aguiar et al. (2019), Hurtado–Nuño–Thomas (2023).
- **No commitment:** DeMarzo–He–Tourre (2023). **Reputation:** Amador–Phelan (2021, 2023), Tsoy (2026).
- **Self-fulfilling crises:** Calvo (1988), Cole–Kehoe (2000), Lorenzoni–Werning (2019), Aguiar et al. (2022).
- **Price floors and spread brakes:** Chatterjee–Eyigungor (2015), Hatchondo–Martinez–Roch (2022).
- **Computation:** Achdou et al. (2022), Rendahl (2022), Mihalache (2025).

The Model

The sovereign

- Preferences $\mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_t)\, dt$. Log endowment z_t, a Markov process.
- Long-term debt B_t: coupon λ, maturity rate δ, debt service $(\delta + \lambda)B_t$.
- Default intensity $d_t \in [0, 1]$: share withheld. A fraction κ becomes arrears.
- Issuance ℓ_t at the market price q_t. Buybacks allowed.

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \qquad \dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t B_t$$

- Write $a(d) := \delta - \kappa(\delta + \lambda)d$: absent issuance, debt decays at rate $a(d)$.
- $1 - \kappa$: the haircut lenders suffer on defaulted payments.

The cost of default

$$\phi(d,z) = (1-\gamma_0 d^{\gamma_1})\, F(z) \;\text{ for } d>0, \qquad \phi(0,z)=1,$$
$$F(z) = 1-\gamma_2\,(e^z - y^*)\,\mathbb{1}\{e^z \geq y^*\}$$

- **Variable cost** $\gamma_0 d^{\gamma_1}$: convex, zero marginal cost at $d=0$.
- **Fixed cost** $(1-F(z))e^z$: paid whenever the sovereign defaults at all.
- Larger in good times. Zero below y^*: the *no-fixed-cost region*.
- The jump of ϕ at $d=0$ creates a region of no default.

Same structure as Arellano, Mateos-Planas, and Ríos-Rull.

Lenders and the bond price

- Competitive, risk-neutral lenders with outside option r. Coupon $\lambda = r$, so $q^{\text{rf}} = 1$.
- A unit pays $(1-d)(\delta+\lambda)$ in cash and $\kappa(\delta+\lambda)d$ in new units.

$$q(B,z) = \mathbb{E}_{B,z} \int_0^\infty e^{-\int_0^t \xi(d_s)\, ds} (1-d_t)(\lambda+\delta)\, dt, \qquad \xi(d) := r + \delta - \kappa(\delta+\lambda)\, d$$

- Feynman–Kac: $\xi(d)\, q = (1-d)(\lambda+\delta) + \tilde{S}\, q_B + \mathcal{L}_z q$, linear in q.
- Spread $= (\delta+\lambda)/q - (\delta+r)$.

Equilibrium

Given the price $q(B,z)$, the value solves

$$\rho V(B,z) = \max_{c,\, d \in [0,1]} \{u(c) + S(B,z,c,d,q)\, V_B + \mathcal{L}_z V\}$$

- Markov equilibrium: V, policies (c,d), and price q that solve both equations.
- A single value function. No exclusion regime, no stopping problem.
- Stationary distribution: the adjoint of the same operator, no simulation.
- First-order condition at an interior drift: $u'(c)\, q = -V_B$.

Arrears are a source of funds at a fixed price

Let n_t be the total addition to debt, from issuance and arrears. Then

$$c_t = \phi(d_t, z_t)e^{z_t} - (\delta + \lambda)B_t + q_t\, n_t + (1 - \kappa q_t)(\delta + \lambda)B_t\, d_t.$$

- Withholding a unit of debt service raises consumption by $1 - \kappa q_t$, at an output cost.
- Its price is fixed by the recovery κ, with the haircut falling on existing creditors.
- Its terms do not deteriorate with the sovereign's creditworthiness.
- So arrears are attractive precisely when market prices are low.

Lenders price expected withholding. No transfer: the output cost is a deadweight loss.

The underwriting standard

- Problem: with no fixed cost, full default makes debt nearly free at the margin.
- Lenders still pay a positive price, so the sovereign borrows up to any limit.
- Quantitative models leave the limit implicit, as the top of the debt grid.

Our fix: underwriters do not place new issues at a price below $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}.$$

- Outstanding debt still trades. Buybacks and arrears unchanged.
- Long-run debt no longer depends on the limit. Binds on $\sim$2% of the mass.
- Full-default bonds are claims on the recovery. A standard above their price suffices.
- A price floor (Chatterjee–Eyigungor 2015), a spread brake (Hatchondo et al. 2022).

The default decision is static

For a given drift S, consumption is $c = \mathcal{R}(d; B, z, q) + q\,S$, with the *resource function*

$$\mathcal{R}(d; B, z, q) := \phi(d, z)e^{z} - (1 - d)(\delta + \lambda)B + q\,a(d)\,B.$$

Proposition

Wherever the standard is slack, every HJB maximizer sets $d^{*} = \arg\max_{d} \mathcal{R}(d; B, z, q)$.

- Independent of the value function and of where debt is going.
- Marginal gain from withholding: $(1 - \kappa q)(\delta + \lambda)B$, payment saved net of arrears.

Explicit thresholds

Interior intensity and the full-default threshold, at any state and price:

$$d_{\text{int}} = \left[\frac{(1-\kappa q)(\delta+\lambda)B}{\gamma_0\gamma_1 F(z)e^z}\right]^{\frac{1}{\gamma_1-1}}, \qquad B_{\text{fd}}(z,q) = \frac{\gamma_0\gamma_1 F(z)e^z}{(1-\kappa q)(\delta+\lambda)}$$

- No default for $B \leq B_{\text{pd}}(z,q)$, also in closed form. Partial default above it.
- B_{pd} rises with the price: a lower price makes withholding cheaper.
- No-fixed-cost region: $B_{\text{pd}} = 0$, any positive debt is partially defaulted on.
- Regions that discrete-time models find only numerically.

A sovereign never defaults a little

- The intensity is a continuous choice, but the fixed cost makes entry lumpy.
- The first default is a jump of a size set by the costs alone,

$$d_{\text{first}}(z) = \left[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\right]^{1/\gamma_1},$$

  about half of the debt service in the calibrated model ($d_{\text{first}} \approx 0.49$, below).
- Independent of debt, the price, and preferences.

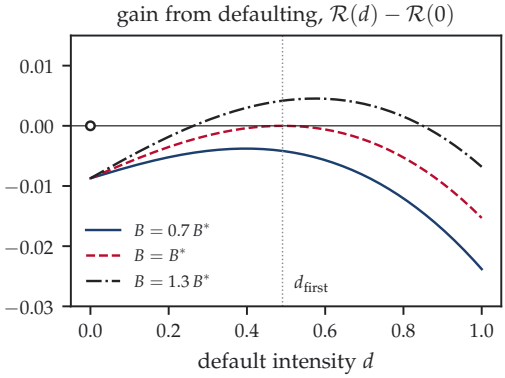
Why the jump

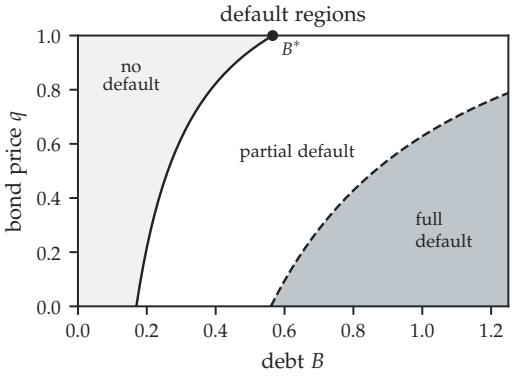
- Without a fixed cost: zero marginal cost at $d = 0$, so a little default at every debt.
- The fixed cost is what makes full repayment possible.
- Once paying the fixed cost, the sovereign withholds until the variable cost bites.
- A steep variable cost (γ_1 large) makes the jump large.

What triggers the jump needs uncertainty

- Not borrowing into default: a shock that moves the threshold below the debt held.
- The chain model adds the one missing element, precaution.

The default regions





- Left: the gain jumps down at $d > 0$, the fixed cost. At B^*, a tie.
- Right: the no-default region shrinks as the price falls. Its edge at q^{rf} is capacity.

The Deterministic Model

The deterministic model

With z fixed at 0, the equilibrium is a pair of ordinary differential equations,

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*},$$

with $d^* = d^*(B, q(B))$ static, $c^* = (u')^{-1}(-V'/q)$, and $S^* = (c^* - \mathcal{R}^*)/q$.

- Singular where the drift vanishes: the steady states organize the solution.
- No-default steady state: debt capacity. Partial-default steady state: self-fulfilling.
- Illustrations use the calibrated parameters with the endowment at its mean.

Debt capacity

Theorem (debt capacity in closed form)
In the deterministic model the impatient sovereign borrows at the risk-free price up to

$$B^* = B_{\text{pd}}(0, q^{\text{rf}}),$$

the largest debt it is willing to service in full at the risk-free price, and stops there.

- Closed form in the default costs, the recovery on arrears, and debt maturity.
- Independent of patience, the curvature of utility, and the borrowing limit.

With shocks: the same closed form, at a price below par.
Preferences enter through that price.

Why the sovereign stops at B^*

- Below B^*: no default, price at par, impatience pushes debt up.
- One more unit past B^* starts a default of size d_{first}.
- Lenders would charge for that default at once: the price leaves par.
- The sovereign prefers to stop exactly at the edge of the no-default region.

Contrast with the binary model

- There, the limit is where staying in the market stops covering repayment.
- It depends on patience, risk aversion, and the punishment after default.
- It is found only numerically.

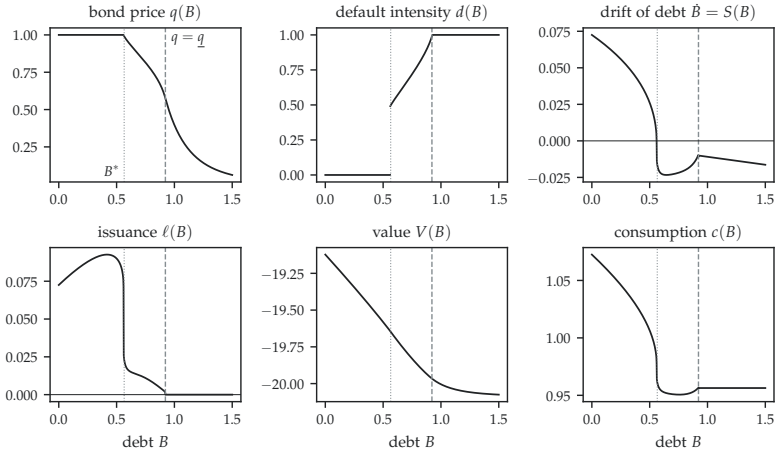
What moves capacity

- **Default costs** γ_0, $1 - F$, and **income**: higher, more capacity.
- **Recovery** κ: more arrears per unit withheld make withholding costlier.
- **Debt service** $\delta + \lambda$: a larger flow to withhold from, lower capacity.

What does not move it

- The discount rate ρ, as long as $\rho > r$. The curvature of u.
- The drift cap and the borrowing limit.

The deterministic equilibrium



- Par price and positive drift up to B^*. Partial default and negative drift beyond it.
- V is C^1 at B^*. Consumption drops steeply but continuously at the first default.

Laws of motion around capacity

Lemma (square-root law, from below)

$$S(B^* - x) = (2(\rho - r)\,\psi(\bar{c}))^{1/2}\, x^{1/2}(1 + o(1)), \qquad \psi = u'/|u''|,$$

and B^* is reached in finite time.

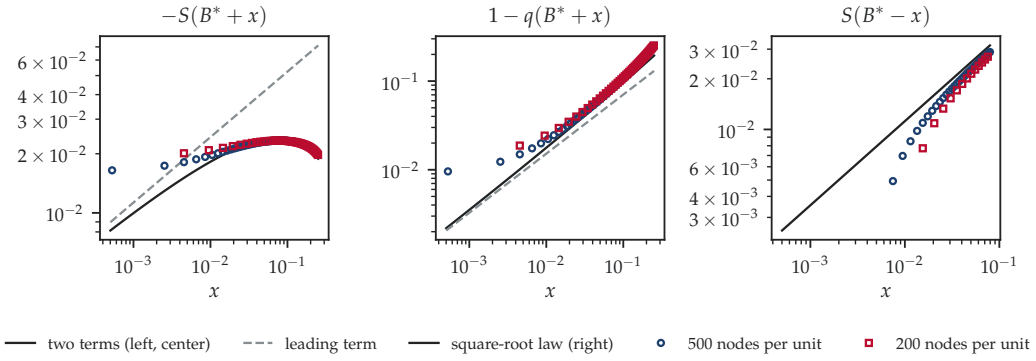
Theorem (cube-root law, from above)

With $\Pi_0 = (1-\kappa)(\delta+\lambda)d_{\text{first}}$ and $\beta = (3a(d_{\text{first}})B^*\Pi_0\,\psi(\bar{c}))^{1/3}$,

$$S(B^* + x) = -\beta\, x^{1/3}(1 + o(1)), \qquad q^{\text{rf}} - q(B^* + x) = \tfrac{3\Pi_0}{2\beta}\, x^{2/3}(1 + o(1)).$$

- The value is C^1 at B^*. Second-order terms in closed form too.

The cube-root law at work



- Repayment slows near capacity. The price leaves par with a vertical tangent.
- Return in finite time: 0.6 years from an excess of 1% of output, 2.9 from 10%.
- Priced at once, worked off without restructuring. The binary model jumps instead.

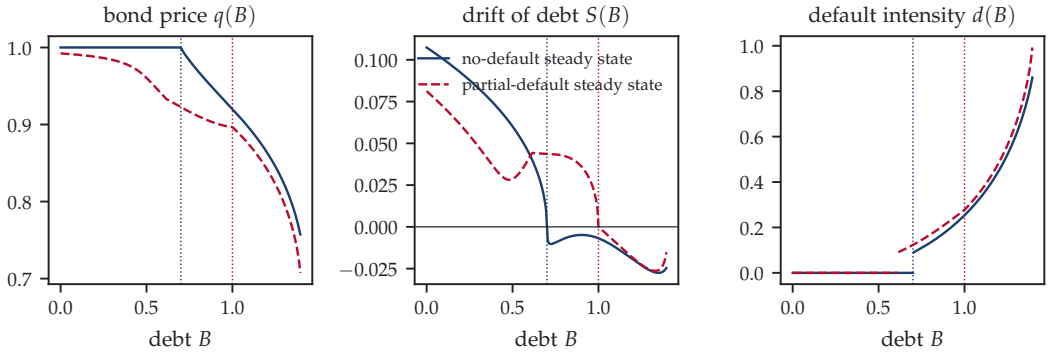
Pessimism alone can produce a slow crisis

- Lenders expect the sovereign to settle at B°, above capacity.
- They expect it to withhold a constant share d° there forever.
- So they pay $q_r(d^\circ) = (1 - d^\circ)(\lambda + \delta)/\xi(d^\circ)$ at B°, and less than par below.

Two consequences validate the expectation

- At the lower price withholding is cheaper, so default starts earlier.
- The default is priced, so the sovereign borrows past its old limit at the risk-free rate.
- It stops at B°, where impatience balances the dilution of its price.

Two equilibria at the same parameters



- Steady states: no default at $B = 0.7$, partial default at $B = 1$ ($d = 28\%$, $q = 0.9$).
- One of a continuum: steady states from 0.93 to 1.02. No run, no loss of access.

When is pessimism self-fulfilling?

Proposition (risk-neutral sovereign)

A self-fulfilling partial default is reached from zero debt only if $\rho - r \geq \Theta^*$: the wedge must cover the output cost of the cheapest default per unit of funds raised. When the default cost is not too convex,

$$\Theta^* = s_{\text{first}},$$

the spread implied by withholding the first default intensity forever. Equilibria exist for a range of wedges above it.

- In the example: $s_{\text{first}} = 47$ bp, stable up to 80 bp.

The discrete-issuance test

- At zero debt, issue the steady-state debt at once and collect its market value.
- From then on: pay the risk-free rate on what was raised, the default is priced.
- And lose output $1 - \phi(d, 0)$ to the default, forever.

Front-loading vs. output cost

- Gain: the impatience wedge times the funds raised, per unit of time.
- Pessimism is self-fulfilling when the gain covers the output cost.
- At the first default, that cost equals the spread times the value of the debt.

Unlike a rollover crisis, the sovereign can stay out of the default region.

Comparison with slow-moving and rollover crises

- **Lorenzoni–Werning (2019):** pessimism raises the cost of rollover, debt drifts.
- **Here:** pessimism raises the default intensity, which justifies the price.
- **Calvo (1988):** expected repudiation raises rates, which justify repudiation.
- **Cole–Kehoe (2000):** a run. Here: no run, no loss of market access.
- **Aguiar et al. (2022):** closest, the government keeps issuing at depressed prices.

The difference: no forced rollover. The sovereign must choose to borrow into default.

The Chain Model

The chain model

The log endowment takes values z_j and switches from j to k at rate η_{jk}.

$$\rho V_j(B) = \max_{c,d} \left\{ u(c) + S\, V_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[V_k(B) - V_j(B) \right] \right\},$$
$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[q_k(B) - q_j(B) \right].$$

- The static default decision and the thresholds hold state by state, at $q = q_j(B)$.
- One new element: precaution. It can hold a steady state short of capacity.
- The bridge between the deterministic model and the diffusion.

Precaution

Endowment on a finite chain, switching from j to k at rate η_{jk}.

Proposition (steady states at the risk-free price)

At an interior steady state B° of state j in a default-free set,

$$(\rho - r)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left(u'(c_k) - u'(c_j) \right).$$

The state with the lowest consumption has no such steady state.

- Impatience gain from borrowing = precautionary loss from a worse state.
- The worst state borrows to its edge. Better states can hold a buffer.

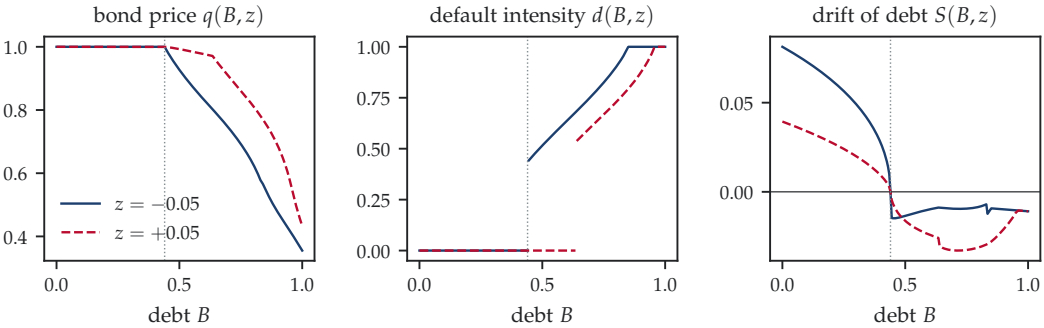
The price kink and when defaults occur

- Past a worse state's threshold, a switch means immediate default: a price kink.
- The kink can hold the better state at or below the worse state's threshold.
- Then a switch does not land the sovereign in default.

Default occurs when a shock moves the threshold below the debt

- Chain: a rare jump. Diffusion: the threshold moves, and the sovereign is caught.
- How often: the tail of shocks against the buffer, not borrowing into default.

A two-state illustration



- $z = \pm 0.05$, switching at rate 0.054. No state defaults on the ergodic set.
- Both rest just below the low state's threshold, 0.445. The high state's own is 0.64.
- The kink holds the high state there: impatience 0.012 against precaution 0.0115.
- More frequent switching (± 0.06, rate 0.15): the high state stops inside, at 0.15.

The Diffusion Model

The diffusion model

Ornstein–Uhlenbeck log endowment, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$:

$$\rho V = \max_{c,\,d} \left\{ u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz} \right\},$$

$$\xi(d)\,q = (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}.$$

- The continuous-time counterpart of the annual AR(1): $\mu = -\ln \rho_z$.
- Stationary density from the adjoint of the same operator.
- Thresholds move continuously with the endowment.

Where the theory shows: regions and capacity in the policies, stationary debt in the ergodic set, the fixed cost in the response to a shock.

Calibration strategy

- Ornstein–Uhlenbeck endowment, $dz = -\mu z\, dt + \sigma\, dW$. CRRA utility, $\nu = 2$.
- Targets and weights of Arellano, Mateos-Planas, and Ríos-Rull.
- Recovery and maturity fixed at their values, which do not depend on the period.
- Seven free parameters: default costs, discount rate, endowment.
- Eleven targets: default frequency and depth, debt, debt service, spreads, output.
- The standard is not a free parameter. It is the lowest level at which the sovereign does not issue in full default on the ergodic set: $\underline{q} = 0.575$.

Parameters

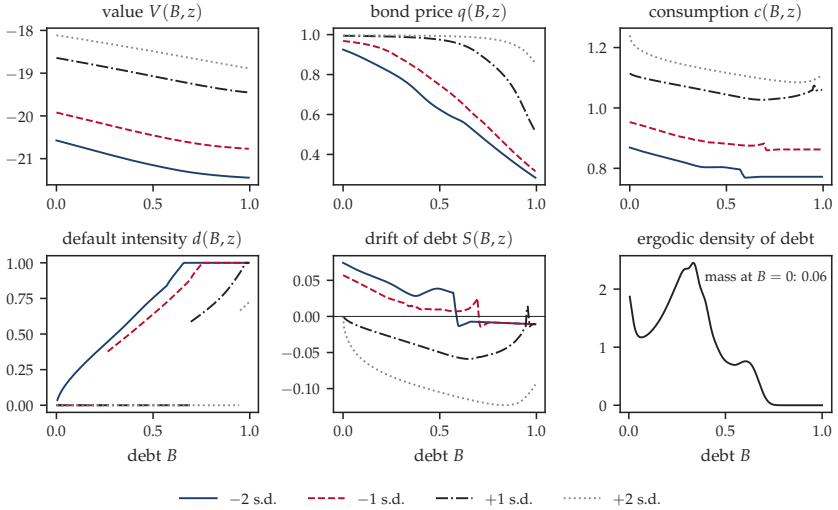
		AMR	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
κ	recovery of defaulted coupons	0.70	0.70
δ	maturity rate ($-\ln 0.88$)	0.128	0.128
Set by a rule			
$\bar{q}$	underwriting standard	–	0.575
Calibrated			
γ_0	variable default cost, level	0.0476	0.0352
γ_1	variable default cost, curvature	2	2.68
γ_2	fixed default cost, slope	0.12	0.044
y^*	fixed default cost, threshold (level)	0.938	0.803
ρ	discount rate (factor $e^{-\rho}$)	0.047	0.052
ρ_z	annual persistence of the endowment	0.875	0.897
σ_ϵ	annual innovation s.d.	0.052	0.047

Fit

Moment	Weight	Data	AMR	Model
Targeted				
Partial default: frequency	1	0.36	0.37	0.39
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.48
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.26
Debt to output: mean	1	0.32	0.32	0.30
Debt to output: s.d.	1	0.18	0.25	0.20
Debt service to output: mean	1	0.036	0.035	0.033
Debt service to output: s.d.	1	0.021	0.022	0.018
Debt due to output: mean	3	0.049	0.050	0.050
Spread: s.d.	1.25	0.041	0.037	0.029
Output: persistence	1	0.89	0.88	0.93
Output: s.d.	1	0.10	0.12	0.10
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.045
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.034
Spread: mean		0.053	0.016	0.020
Spread: correlation with output		−0.17	−0.38	−0.76
Spread: correlation with debt		0.24	0.56	0.84
Consumption s.d. relative to output s.d.		1.00	0.91	0.93

Matches default frequency, debt, debt service, debt due, and output volatility.
Deeper default and a lower spread s.d. than the data. Loss 0.260, against their 0.228.

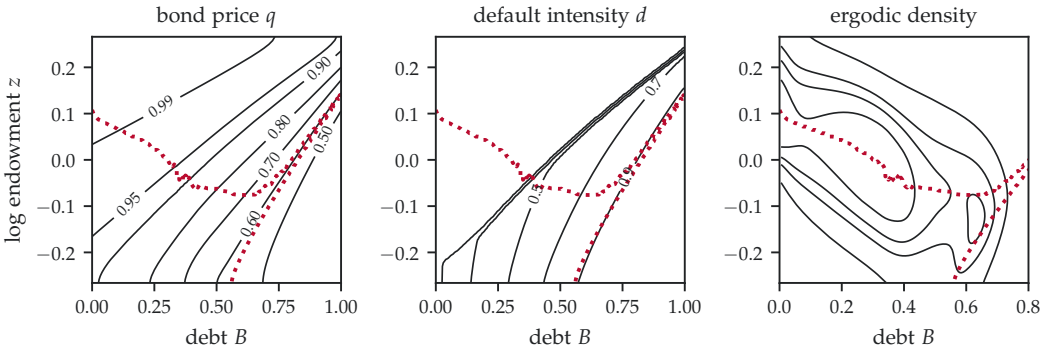
Values, policies, and prices



Reading the policies

- The default policy follows the closed-form rule at every endowment.
- Lowest endowment (no fixed cost): default at any positive debt.
- Average and high endowments: no default until $B_{\text{pd}}(z, q)$, then a jump to d_{first}.
- The threshold rises with the endowment. Full default on $\sim$0.1% of the mass.
- Consumption falls at the threshold, nearly flat in debt beyond it.
- Spreads low most of the time, high in the tail.

The ergodic set



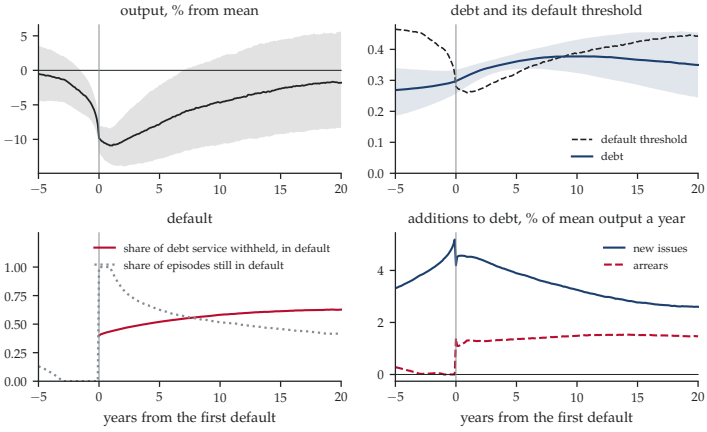
- No-default region: a wedge in (B, z). Mass concentrates along the zero-drift locus.

Stationary debt across states

- **Good times:** stationary debt below the no-default threshold, held by precaution.
- **Bad times:** threshold below the debt. Partial default, fixed cost sunk.
- Stationary debt is set by the dilution of its own price, inside the default region.
- **Worst states:** stationary debt where the price reaches the standard.
- **Very good times:** the sovereign repays all the way to zero debt.

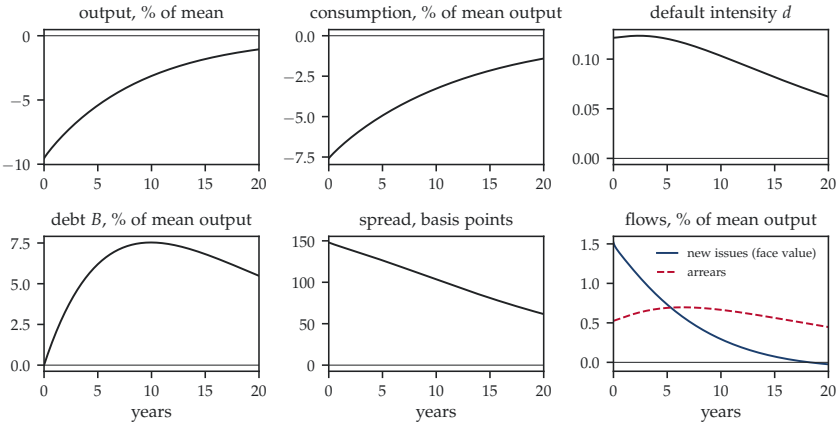
The locus rises from good to bad times, peaks at $B \approx 0.73$, and bends back to 0.56. Default is lumpy at entry and sticky thereafter.

The typical default episode



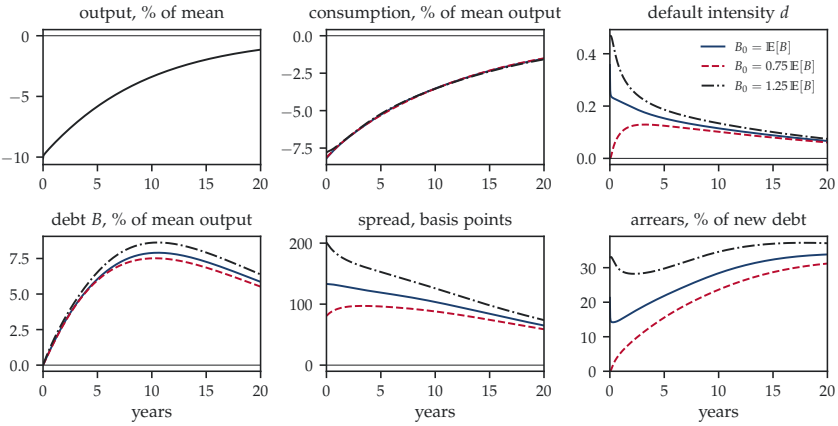
- Output ∼10% below its mean at entry. The threshold falls through the debt: ∼40% withheld at once.
- Median debt rises for ten years, by a quarter. Half of the episodes are over within six years.

After a bad shock



- To the peak, issues add 7.9% of output to the debt, arrears 6.5%. Issues are larger for 5.3 years.
- The episode ends when output recovers, not when the debt is repaid.

The same shock from three states



- Debt at its ergodic mean, 25% below, and 25% above. Debt responds the same way from all three.
- Default and the spread depend on the state. Arrears: at most a third of gross new debt.

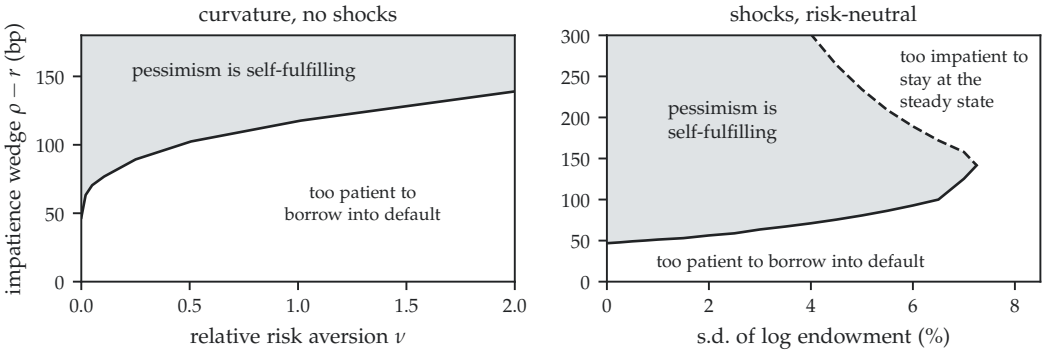
What the episode explains

- Default jumps on impact and stays elevated for two decades.
- Debt rises for about ten years, by 7.5% of output.
- The spread rises by about 148 bp on impact and declines with recovery.
- New issues are the larger flow for 5.3 years, arrears thereafter.

The facts of partial default episodes

- Default is protracted. Debt rises during the episode.
- Countries exit owing more than when they entered.
- New issues and arrears add about equally to the debt up to its peak.

Curvature and uncertainty raise the bar



- Risk aversion of two roughly triples the impatience needed (139 vs. 47 bp).
- Shocks: window narrows from both sides, closes at endowment s.d. $\approx 7\%$.

Why curvature and shocks work against it

Curvature

- A sovereign that values smooth consumption gains less from front-loading.
- It cannot make a discrete issuance. It drifts into default gradually.
- The threshold rises with risk aversion and approaches s_{first} only slowly.

Uncertainty (risk-neutral, steady debt independent of the endowment)

- From below: in bad states, a deeper and costlier default at the steady state.
- From above: the steady state must be stable in every state.
- Waiting at zero debt for a better endowment does not help.

Self-fulfilling default at the calibrated parameters

- Stationary spread at the first default: 483 bp. Spread at the standard: 1,235 bp.
- So the standard does not exclude partial-default steady states. Impatience does.
- Impatience wedge: 128 bp, against a risk-neutral threshold of 483 bp.
- Curvature and shocks raise the bar further.
- A second equilibrium exists: default on the last 3.1% of borrowing, up to 43 bp.

The calibrated sovereign defaults because of shocks, not because of pessimism.

Conclusion

Takeaways

- Arrears raise funds at a price fixed by the recovery. The haircut falls on creditors.
- Debt capacity is a closed form in the cost of defaulting and the bond price.
- Crises begin with a jump, when shocks overwhelm a precautionary buffer.
- The episode is a financing regime: borrowing continues until output recovers.
- Pessimism can cause a slow crisis without a run. Curvature, shocks raise the bar.

Tractable and quantitative at once: fits AMR's targets about as well as their model.

Robustness

Robustness of the calibrated point

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Unresolved
main grid	0.300	0.204	0.029	0.389	0.477	0.259	0.029
debt step halved	0.300	0.203	0.029	0.388	0.477	0.260	0.011
non-uniform debt grid	0.300	0.203	0.029	0.389	0.478	0.261	0.008
limit at $\bar{B} = 1.5$	0.300	0.204	0.029	0.389	0.477	0.259	0.028
$n_z = 51$	0.301	0.205	0.030	0.395	0.474	0.261	0.034
$n_z = 101$	0.300	0.203	0.029	0.387	0.478	0.259	0.025
endowment range ± 3 s.d.	0.300	0.206	0.030	0.390	0.484	0.267	0.026
endowment range ± 3.5 s.d.	0.300	0.205	0.030	0.391	0.483	0.265	0.028

- Finer, non-uniform, and wider debt grids, fewer and more endowment nodes: moments and loss unchanged.
- Without the standard the moments depend on the limit: spread s.d. 0.053 on $[0, 1]$, 0.056 on $[0, 2]$.
- Wider endowment range: debt, default, and spread moments unchanged. The loss rises with the s.d. of output.

Other Material

Computation: the scheme

- Implicit upwind finite differences (Achdou et al. 2022), 201×75 grid on (B, z).
- Value and price updated jointly each step. Alternating between them is unstable.
- At each node, three candidates: forward, backward, and zero drift.
- For each: closed-form drift, golden-section search over d, plus $d \in \{0, 1, d_{\text{int}}\}$.
- Keep the candidate with the largest Hamiltonian: monotone upwinding.
- Handles the fixed cost's non-concavity directly. No taste shocks.

Computation: accuracy

- Residuals of the main solution: HJB 2×10^{-9}, pricing 9×10^{-9}.
- The standard takes effect over a band of prices, so the policies are continuous where it binds.
- Stationary distribution: the transposed generator, solved directly.
- Impulse responses: the forward equation on the shifted distribution. No sampling.
- Check: an exact reverse-shooting solution of the deterministic discrete system.

High debt and the underwriting standard

- Without a limit, debt cannot stay above every full-default threshold forever.
- With a limit and no standard, the sovereign borrows up to it, whatever its level.
- With the standard: debt shrinks in full default, the full-default region is transient.
- A bond in full default until the crisis ends is worth at most $q_j^* = q^{\text{rf}}\,\eta_j/(r+a+\eta_j)$.
- A standard above q_j^* rules out the accumulation, in every equilibrium.
- Where it binds, default follows the closed-form rule at the shadow price $-V_B/u'(c)$.
- Highest price in full default on the ergodic set: 0.54, below the standard of 0.575.

The price and the value

- Price in $[0, q^{\mathrm{rf}}]$: q^{rf} only on default-free paths, 0 only in full default forever.
- The price can jump only where debt stops and is not reached from both sides.
- Full default forever, at a zero price: a knife edge, ruled out by no trade at $q = 0$.

Local conditions and the continuum

Lemma (at a partial-default steady state)
The price is $q^\circ = q_r(d^\circ)$, and the slope of the price is pinned by impatience,

$$-\frac{q'(B^{\circ+})}{q^\circ} = \frac{\rho - r}{a(d^\circ)\, B^\circ}.$$

- One candidate for every intensity $d^\circ \geq d_{\text{first}}$: a continuum.
- Stable only if more debt dilutes the price enough, through more default.
- Equilibrium depends on the whole path: borrowing into default from zero debt.
- Lenders' expectations alone select among the candidates.

Self-fulfilling default with shocks: the construction

Risk-neutral sovereign, diffusion endowment, expected steady debt B° in every state.

- At B°: $\xi(d^\circ)q^\circ = (1-d^\circ)(\lambda+\delta) + \mathcal{L}_z q^\circ$, an ODE in z.
- Above B° the sovereign is indifferent, $V_B = -q$, and the price solves

$$a(d)B\,q_B = (1-d)(\lambda+\delta) - (\rho + a(d))\,q + \mathcal{L}_z q.$$

- Below B°: a discrete issuance at q°. It must beat staying debt-free in every state.
- Checks: stability, continuation, the standard. At $\sigma = 0$: the deterministic test.